

Yun Chen

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EDUCATION

- **University of California San Diego (UCSD)** San Diego, CA
Ph.D., Electrical and Computer Engineering (Degree Conferred) Apr. 2024 - Apr. 2025
- **North Carolina State University (NCSU)** Raleigh, NC
Ph.D. Program, Electrical and Computer Engineering (GPA: 4.0/4.0) – transferred to UCSD Jan. 2021 - Mar. 2024
- **The University of Texas at Austin (UT Austin)** Austin, TX
M.S., Electrical and Computer Engineering (GPA: 3.95/4.0) Aug. 2018 - Dec. 2020
- **Southeast University** Nanjing, China
Bachelor of Engineering in Information Engineering (GPA: 89.13/100) Aug. 2013 - Jun. 2017

RESEARCH AND INDUSTRY EXPERIENCE

Industry Experience

- **Senior Engineer** Apr. 2025 - Present
Qualcomm Technologies, Inc.
 - Full-time engineering role, building on Ph.D. research and prior Qualcomm internships
- **Engineering Intern** Summers 2021, 2022 & 2023
Qualcomm Technologies, Inc.
 - ML-based detection and localization for indoor device-free RF sensing – Dr. Marwen Zorgui, Dr. Arumugam Kannan (2023)
 - Precise indoor positioning using semi-supervised learning – Dr. Srinivas Yerramalli (2022)
 - ML-assisted coordinated multi-point transmission for ultra-reliable services – Dr. Taesang Yoo (2021)
- **Research Intern** Summers 2019 & 2020
Ericsson Inc. – Prof. Ali Khayrallah
 - Deep learning-based 5G air-to-ground network design and optimization [12.][14.] (2020)
 - Reinforcement-learning-based drone mobility and handover management [15.][17.] (2019)

Academic Research

- **Graduate Researcher** Sep. 2019 - Mar. 2025
WiSeCom Lab / UT Austin, NC State University, UC San Diego – Prof. Nuria González Prelcic
 - Hybrid model/data-driven mmWave localization, channel estimation, and user tracking for ISAC systems
 - Attention-network-based vehicle position and channel tracking at mmWave [2.][7.][8.]
 - Collaborative V2V sidelink localization for orientation estimation and position tracking [4.]
 - Low-complexity mmWave channel estimation via ESPRIT-D [6.]
 - Radar-to-communication channel mapping for mmWave link configuration [9.][10.][13.]
 - UAV navigation using a monocular camera and deep reinforcement learning [16.]
- **Undergraduate Research Assistant** Jan. 2017 - Jun. 2017
Southeast University – Prof. Luxi Yang, Prof. Chunguo Li
 - Trajectory optimization for bicycle trip planning using reinforcement learning [18.]
 - Proactive caching in wireless networks using transfer learning [19.]
 - Mobile edge computing resource allocation using an Echo State Network

PATENTS

- X. Lin, **Y. Chen**, M. Mozaffari, T. Khan, “Efficient 3D Mobility Support Using Reinforcement Learning”. US20240056933A1. Feb. 2024
- M. Mozaffari, X. Lin, T. Khan, M. Afshang, **Y. Chen**, “Network Design and Optimization Using Deep Learning”. US20230388196A1. Nov. 2023

PUBLICATIONS

1. **Y. Chen**, “Hybrid Model/Data-Driven Solutions in ISAC: Improving Link Establishment, Channel Estimation, and User Localization in mmWave Vehicular Networks”, Ph.D. Dissertation, University of California San Diego, Advisor: Prof. Nuria González-Prelcic, 2026. ProQuest Dissertations & Theses Global, Publication No. 32738029.
2. **Y. Chen**, N. González-Prelcic, T. Shimizu, C. Mahabal, “A Hybrid Model/Data-Driven Solution to Channel, Position, and Orientation Tracking in mmWave Vehicular Systems”, *IEEE Journal on Selected Areas in Communications*, vol. 44, pp. 927–941, 2026.
3. S. Javid, **Y. Chen**, N. González-Prelcic, “Learning-Based Cross-Frequency Channel Prediction in the Upper Mid-band”, in *IEEE Global Communications Conference (GLOBECOM 2025)*, pp. 6394–6399, Dec. 2025.
4. **Y. Chen**, N. González-Prelcic, T. Shimizu, C. Mahabal, “Collaborative Position and Orientation Tracking in mmWave Vehicular Systems Exploiting V2I and V2V Links”, in *2024 58th Asilomar Conference on Signals, Systems, and Computers*, pp. 407–412, Oct. 2024.
5. B. Kumar, **Y. Chen**, N. González-Prelcic, T. Shimizu, A. Ganlath, “DeepSatLoc: A Multimodal Fusion Strategy for Enhanced Localization in Urban Canyons Exploiting GPS and LEO Satellite Communication Signals”, in *2024 58th Asilomar Conference on Signals, Systems, and Computers*, pp. 842–849, Oct. 2024.
6. **Y. Chen**, N. González-Prelcic, T. Shimizu, H. Lu, C. Mahabal, “Beamspace ESPRIT-D for Joint 3D Angle and Delay Estimation for Joint Localization and Communication at MmWave”, in *IEEE International Conference on Communications (ICC 2024)*, pp. 1915–1920, Jun. 2024.
7. **Y. Chen**, N. González-Prelcic, T. Shimizu, H. Lu, “Learning to Localize with Attention: from sparse mmWave channel estimates from a single BS to high accuracy 3D location”, *arXiv:2307.00167*, Jun. 2023, revised Mar. 2025.
8. **[Best Student Paper] Y. Chen**, N. González-Prelcic, T. Shimizu, H. Lu, C. Mahabal, “Sparse Recovery with Attention: A Hybrid Data/Model Driven Solution for High Accuracy Position and Channel Tracking at mmWave”, in *2023 IEEE 24th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, pp. 491–495, Sep. 2023.
9. N. González-Prelcic, A. Ali, **Y. Chen**, “Radar-Aided Millimeter Wave Communication”, in *Signal Processing for Joint Radar Communications*, Wiley-IEEE Press, pp. 155–177, Mar. 2024.
10. A. Graff, **Y. Chen**, N. González-Prelcic, T. Shimizu, “Deep Learning-Based Link Configuration for Radar-Aided Multiuser mmWave Vehicle-to-Infrastructure Communication”, *IEEE Transactions on Vehicular Technology*, vol. 72, no. 6, pp. 7454–7468, Jun. 2023.
11. **Y. Chen**, J. Palacios, N. González-Prelcic, T. Shimizu, H. Lu, “Joint Initial Access and Localization in Millimeter Wave Vehicular Networks: a Hybrid Model/Data Driven Approach”, in *2022 IEEE 12th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, pp. 355–359, Jun. 2022.
12. G. Geraci, A. Garcia-Rodriguez, M. Azari, A. Lozano, M. Mezzavilla, S. Chatzinotas, **Y. Chen**, S. Rangan, M. Renzo, “What Will the Future of UAV Cellular Communications Be? A Flight From 5G to 6G”, *IEEE Communications Surveys & Tutorials*, vol. 24, no. 3, pp. 1304–1335, 2022.
13. **Y. Chen**, A. Graff, N. González-Prelcic, T. Shimizu, “Radar Aided mmWave Vehicle-to-Infrastructure Link Configuration Using Deep Learning”, in *2021 IEEE Global Communications Conference (GLOBECOM)*, pp. 1–6, Dec. 2021.
14. **Y. Chen**, X. Lin, T. Khan, M. Afshang, M. Mozaffari, “5G Air-to-Ground Network Design and Optimization: A Deep Learning Approach”, in *2021 IEEE 93rd Vehicular Technology Conference (VTC2021-Spring)*, pp. 1–6, Apr. 2021.
15. **Y. Chen**, X. Lin, T. Khan, M. Mozaffari, “A Deep Learning Approach to Efficient Drone Mobility Support”, in *Proceedings of the 2nd ACM MobiCom Workshop on Drone Assisted Wireless Communications for 5G and Beyond*, pp. 67–72, Sep. 2020.
16. **Y. Chen**, N. González-Prelcic, R. W. Heath, “Collision-Free UAV Navigation with a Monocular Camera Using Deep Reinforcement Learning”, in *2020 IEEE 30th International Workshop on Machine Learning for Signal Processing (MLSP)*, pp. 1–6, Sep. 2020.
17. **[Best Paper] Y. Chen**, X. Lin, T. Khan, M. Mozaffari, “Efficient Drone Mobility Support Using Reinforcement Learning”, in *2020 IEEE Wireless Communications and Networking Conference (WCNC)*, pp. 1–6, May 2020.
18. **Y. Chen**, W. Yan, C. Li, Y. Huang, and L. Yang, “Personalized Optimal Bicycle Trip Planning Based on Q-learning Algorithm”, in *2018 IEEE Wireless Communications and Networking Conference (WCNC)*, pp. 1–6, Apr. 2018.
19. Y. Wang, **Y. Chen**, H. Dai, Y. Huang, and L. Yang, “A Learning-Based Approach for Proactive Caching in Wireless Communication Networks”, in *2017 9th International Conference on Wireless Communications and Signal Processing (WCSP)*, pp. 1–6, Oct. 2017.

TEACHING EXPERIENCE

- **Teaching Assistant** Winter 2025
Circuits and Systems, UC San Diego - Prof. Robert Heath
- **Teaching Assistant** Spring 2019
Probability and Random Process, UT Austin - Prof. Pedro Santacruz

HONORS AND AWARDS

- Best Student Paper Award in *IEEE SPAWC 2023* Sep. 2023
- Summer Graduate Fellowship of the College of Engineering at NCSU May 2022
- Best Paper Award in *IEEE WCNC 2020* (Top 4 out of 400 accepted papers) May 2020
- Excellent Graduation Project at SEU (Top 10 out of 260 projects) Jun. 2017
- Honorable Mention in the Mathematical Contest in Modeling (MCM) Apr. 2016
- Leike Scholarship at SEU (Top 5% out of 1600 students) Jun. 2015
- 2nd Prize out of 1,000 students in Provincial Advanced Mathematics Competition Jun. 2014

PROFESSIONAL SKILLS

- **Skills:**
 - Programming languages: Python, Matlab, C++, HTML
 - Platforms: PyTorch, Tensorflow
 - Simulation tools: Wireless Insite, Robot Operating System (ROS)
- **Language:** English (fluent), Mandarin (native)